

Where climate enters a credit model

The question

Where does climate actually enter a credit model? Here it enters through one channel, written as six steps of arithmetic that a reviewer can check one by one.

The estimand

The change in the one-year probability of default of the EU industry sector (NACE B-E) in 2040 under the NGFS Delayed transition scenario relative to Net Zero 2050, through carbon cost to profit share to log-odds of default. Static sector structure: no abatement, no demand response, no second-round effects. Positive: the disorderly scenario carries the higher PD.

The answer

-9.5% (-23.7 bp at a 2% baseline PD), literature range [-14.2%, -5.5%]. The sign is negative because Delayed transition has a lower carbon price than Net Zero 2050 in every year from 2025 to 2050 in all three models. Against today, the channel lifts industry PD by +24.5% under Net Zero 2050 and +12.6% under Delayed transition.

Read this first

The magnitude is not established. The mapping from profit to default comes from a US equity study; on European bankruptcy data it fails out of sample (calibration slope 0.18, interval [-0.07, 0.44]; 1 would be calibrated). Rescaled by that slope the headline is -1.8% [-4.3%, +0.7%]. The sign flips if the orderly reference is Below 2°C (+9.3%) or if industry abates with the region (+2.1%).

Data

NGFS Phase 5 (IIASA Scenario Explorer, 294 rows), Eurostat national accounts, air emissions accounts, capital stocks and bankruptcy declarations (1,176 country-sector-year changes), and the coefficient table of Campbell, Hilscher and Szilagyi (2008), parsed from the paper by code. Fetched 2026-10-08. Reference year of the ratios: 2022.

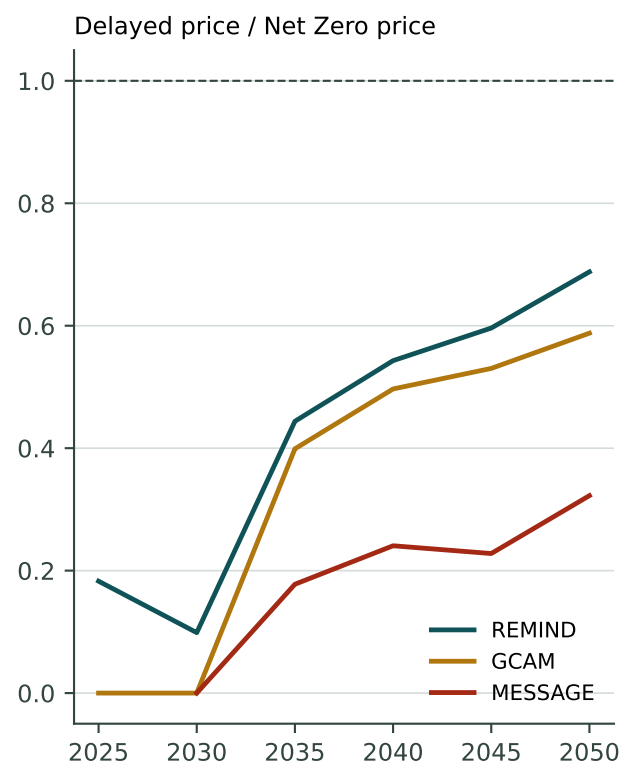
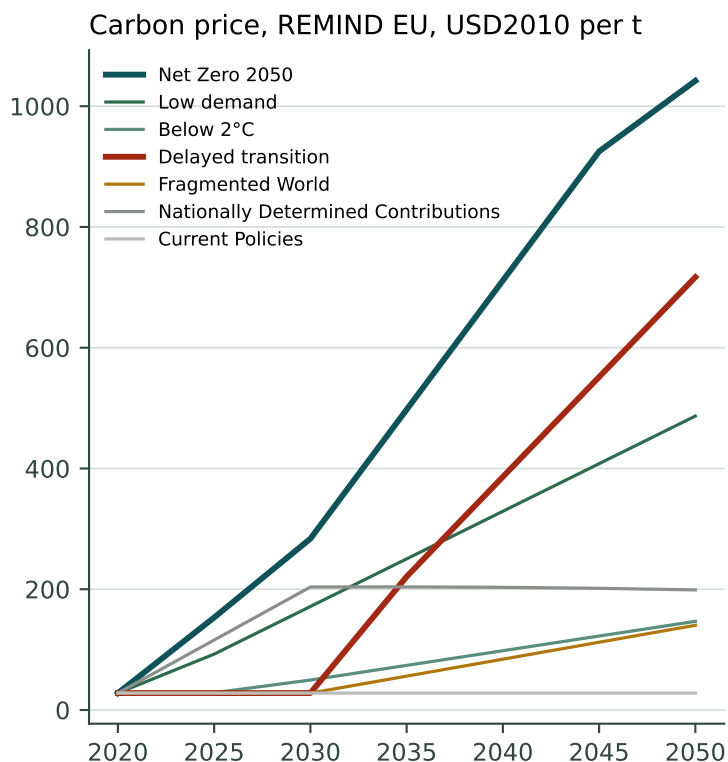
Method: six steps, each one line

The chain, with the numbers at each step

- 1 Price 2040, REMIND EU: Net Zero 2050 US\$711, Delayed transition US\$386 per t (US\$2010).
- 2 Restated in euro of 2022: $\times 1/1.3257$ (2010 rate) $\times 1.254$ (HICP): EUR 673 and EUR 365.
- 3 Cost share of value added = price $\times 518$ t CO₂ per EUR mn GVA: 34.8% and 18.9%.
- 4 Profit share falls by $(1 - 0.7)$ of that. Industry profit share today: 51.5%.
- 5 Quarterly net income over assets falls by that $\times$ GVA/assets $(0.423) / 4$; log-odds rise by -20.26 times it: $+0.224$ and $+0.122$.
- 6 PD from a 2% base: 2.49% under Net Zero 2050, 2.25% under Delayed transition.

Why the sign is what it is

In REMIND, Delayed transition holds the price at US\$28 through 2030 and reaches US\$386 in 2040; Net Zero 2050 is at US\$284 in 2030 and US\$711 in 2040. The delay does not catch up inside the horizon. Only the cost channel is in this module; the abruptness that makes a transition disorderly acts through demand, asset prices and financing conditions, none of which this channel carries.



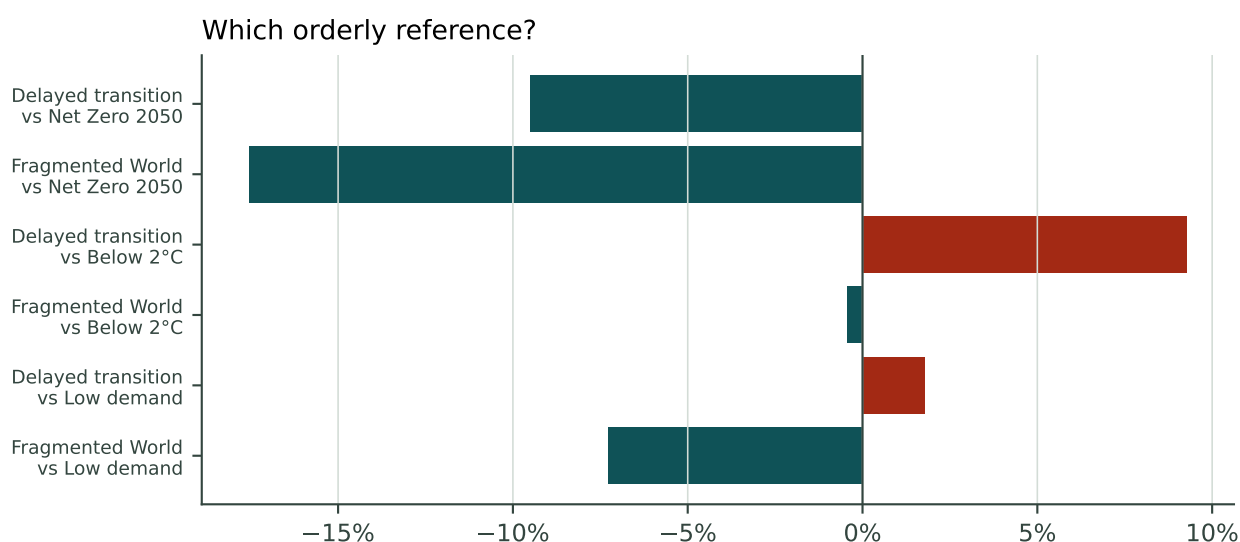
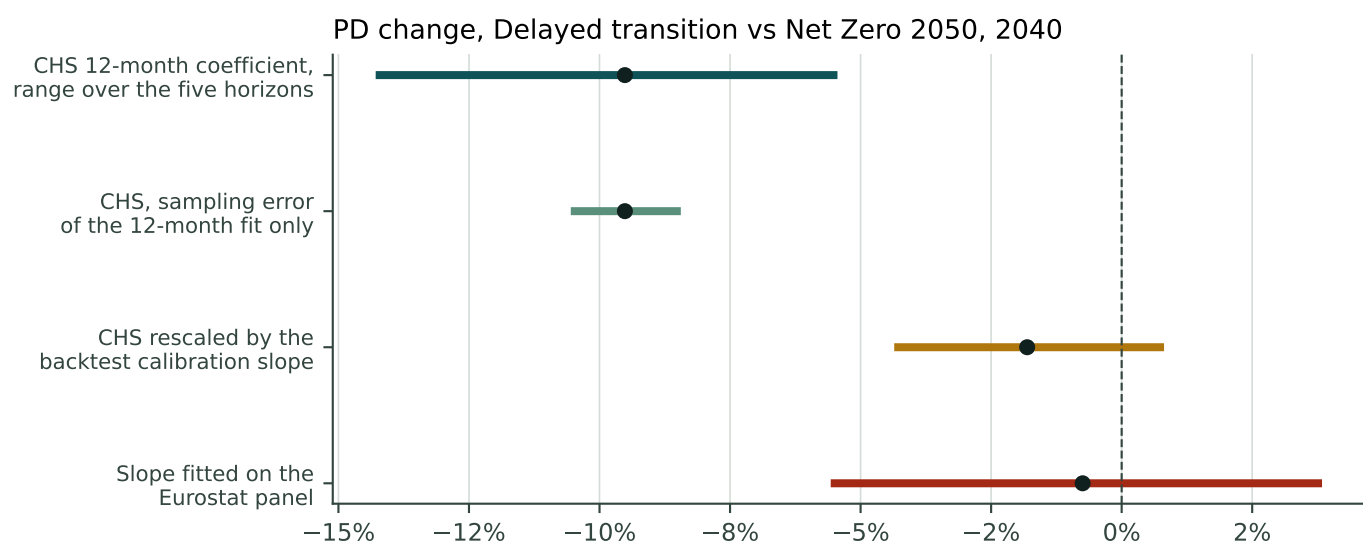
Result: the PD range the elasticity implies

What the four mappings give

Headline -9.5%. The literature sensitivity is not one number: the five horizons in the paper's Table 3 run from -29.7 to -13.2, and drawing one of them and a normal error around it gives [-14.2%, -5.5%]. The sampling error of the 12-month fit alone is [-10.5%, -8.5%], which is the interval one would quote if the US equity result transferred. My own fit on Eurostat bankruptcy declarations gives -0.0016 log points per percentage point of profit share, interval [-0.0121, 0.0079], which includes zero: the PD change is -0.7% [-5.5%, +3.8%].

The scenario pair matters more than the elasticity choices

Delayed transition is below Net Zero 2050 but above Below 2°C in the REMIND region at 2040, so the sign depends on which NGFS scenario is called orderly: the six pairs run from -17.6% to +9.3%, a span of 26.8% points against at most 9.6% for the elasticity choices (source, horizon, asset scaling).



Where in the economy the channel lands

Concentration

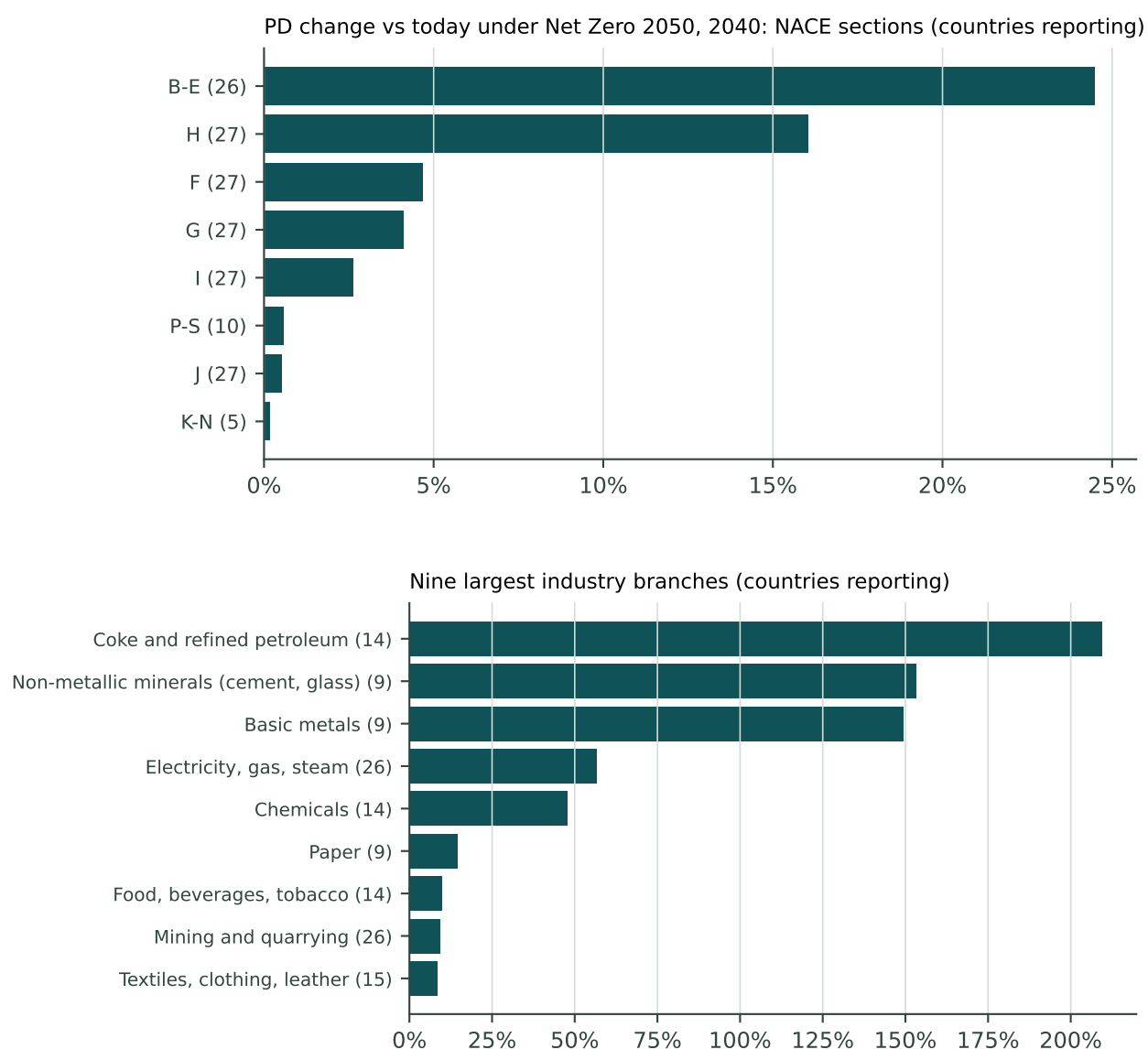
Industry as a whole carries 518 t CO₂ per EUR mn of value added; transport (H) carries 580, and every other section 67 or less. Within industry, coke and refined petroleum, non-metallic minerals (cement, glass) and basic metals take the largest hits (+209%, +153%, +149% against today under Net Zero 2050); the industry aggregate of +24% is an average over a distribution that starts near zero.

Order of sectors

The ordering of the eight sections by shock is stable across reference years 2015 to 2023: the lowest rank correlation with the 2022 ordering is 0.95. It does not depend on the elasticity or the pass-through, which scale every sector alike.

Coverage

Branch figures use the countries that report every series for that branch, so branches rest on 9 to 26 countries; K-N and P-S rest on 5 and 10 and carry little carbon cost.



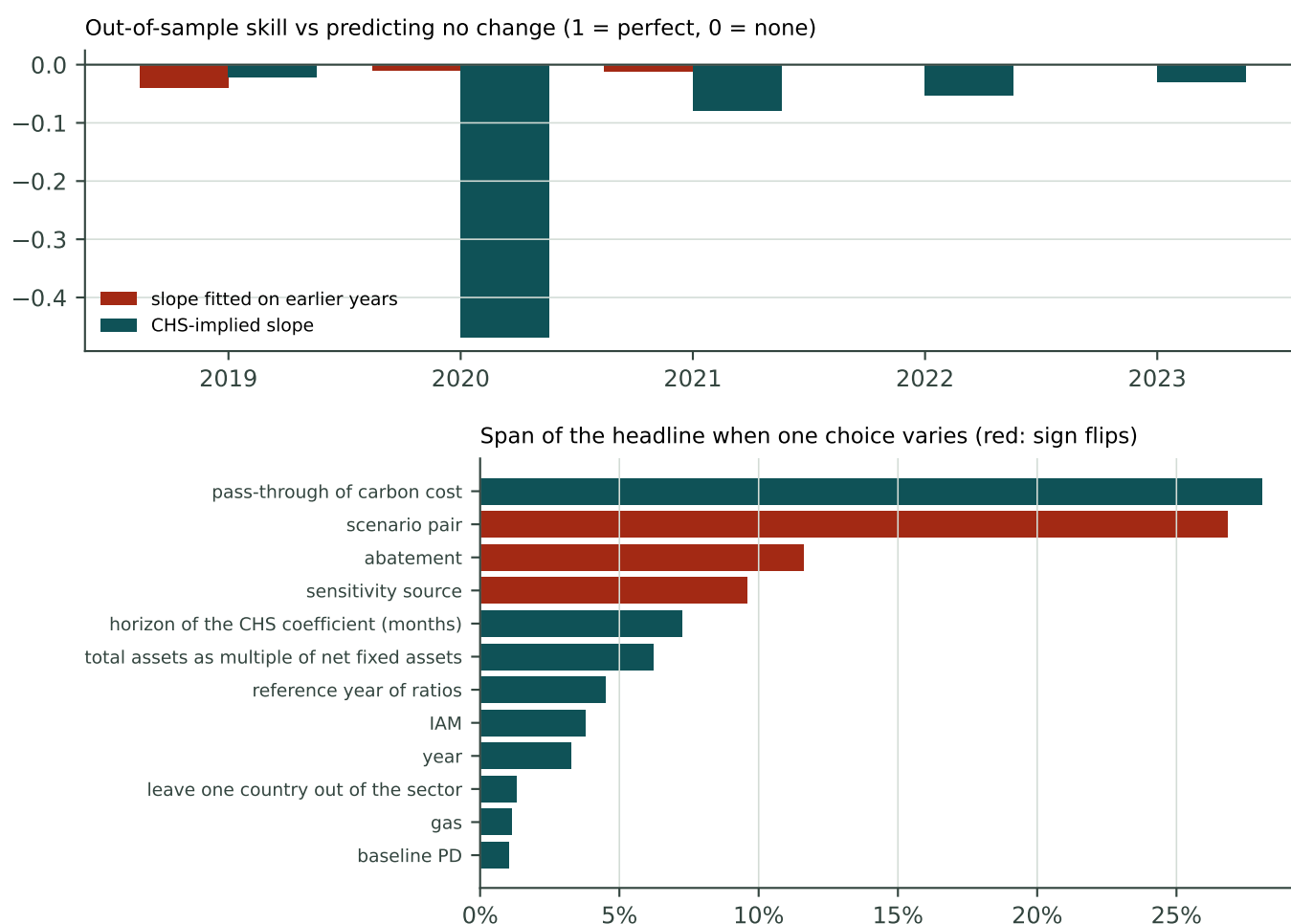
Validation and sensitivity

Backtest of the mapping

For each test year from 2019 to 2023 I fit the profit-to-bankruptcy slope on earlier years only and predict that year's sector-year-demeaned change in bankruptcy declarations from the change in profit share. Over 736 country-sector-years the fitted slope has skill -0.009 and the CHS-implied slope -0.100 against a no-change forecast (RMSE 0.348). The calibration slope of realised on CHS-implied is 0.18 [-0.07, 0.44]; excluding 2020 and 2021, 0.31 [-0.01, 0.71]. Bankruptcy declarations are not PDs, and noise in sector profit share attenuates any slope, so this bounds the mapping from below and does not refute it.

What moves the headline

Largest spans: pass-through of carbon cost (-28.1% to +0.0%), scenario pair (-17.6% to +9.3%), abatement (-9.5% to +2.1%). Leaving out any one country moves it only between -9.8% and -8.5%.



Limits, and what I would build next

The assumption that breaks the conclusion

That a carbon bill passed through a static industry profit share reaches default through a sensitivity estimated on US listed firms. If European default responds to profit as weakly as the bankruptcy backtest suggests, every PD change here shrinks by 81% (rescaled headline -1.8%); if industry abates with the region, the Net Zero 2050 scenario's regional emissions are -95 Mt CO₂ in 2040 against 3325 in 2020, its bill shrinks with them and the sign of the headline reverses (+2.1%). A reviewer should attack the mapping first, then the static structure.

Other limits

- Pass-through is 0.7, from the 70% that Ganapati, Shapiro and Walker (2020) report for US manufacturing energy costs; it is an assumption here, swept from 0 to 1, and it moves the headline from -28.1% to +0.0%.
- Net fixed assets stand in for the market value of total assets; grossing them up by two or three shrinks the effect.
- CHS estimate a monthly failure hazard on quarterly profit; I treat the log-odds slope as transferable to a one-year PD.
- The baseline PD of 2% is a stated parameter and not data: no public source gives sector PDs. The relative change hardly depends on it; the basis-point figure scales with it.
- NGFS prices are model outputs for one region each; nothing here checks them against observed carbon prices.

Next

A bank's own sector default series would replace the bankruptcy proxy and test the elasticity directly. A second step adds the demand side (output response to price) and the abatement cost, so that abatement does not make the carbon bill vanish.